Weekly Assignment: Combinations and the Binomial Theorem (Epp Ch. 9)

Complete each of the following problems. Show all work and justify your answers.

Problem 1 (Epp 9.5 #2).

a. List all 3-combinations for the set $\{x_1, x_2, x_3, x_4, x_5\}$. Deduce the value of $\binom{5}{3}$.

b. List all unordered selections of two elements from the set $\{x_1, x_2, x_3, x_4, x_5, x_6\}$. Deduce the value of $\binom{6}{2}$.

Problem 2 (Epp 9.5 #4). Write an equation relating $P(8, 3)$ and $\binom{8}{3}$.

Problem 3 (Epp 9.5 #7). A computer programming team has 13 members.

a. How many ways can a group of seven be chosen to work on a project?

b. Suppose seven team members are women and six are men.

   (i) How many groups of seven can be chosen that contain four women and three men?
   (ii) How many groups of seven can be chosen that contain at least one man?
   (iii) How many groups of seven can be chosen that contain at most three women?

c. Suppose two team members refuse to work together on projects. How many groups of seven can be chosen to work on a project?

d. Suppose two team members insist on either working together or not at all on projects. How many groups of seven can be chosen to work on a project?

Problem 4 (Epp 9.6 #4). A camera shop stocks eight different types of batteries, one of which is type A76. Assume there are at least 30 batteries of each type.

a. How many ways can a total inventory of 30 batteries be distributed among the eight different types?

b. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory must include at least four A76 batteries?

c. How many ways can a total inventory of 30 batteries be distributed among the eight different types if the inventory includes at most three A76 batteries?

Problem 5 (Epp 9.7 #12). Use Pascal's formula repeatedly to derive a formula for $\binom{n+3}{r}$ in terms of values of $\binom{n}{k}$ with $k \le r$. (Assume n and r are integers with $n \ge r \ge 3$.)